From Growth Model to Planning Assistant: Terrain-Conditioned Diffusion with Auditable Language Operations and Jurisdictional Rule Packs

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Code: https://github.com/eshankiyer/terrain-urban-diffusion

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Abstract

Municipal planning consists of many distinct duties, and only some of them are automatable. This paper presents a pipeline that covers a defined subset of those duties end to end and states which duties it does not touch. At the core is a conditional denoising diffusion model that generates street-level growth alternatives (roads, continuous built-up density, and proposed amenity locations) for real settlements, trained on observed 1980–2020 change assembled automatically from OpenStreetMap, open elevation tiles, and the multi-epoch GHS-BUILT-S surface. Growth style is provided by a mixture of five experts, one per contemporary growth regime (village, hilly town, planned fringe, informal peri-urban frontier, and megacity edge), dispatched by a hard, auditable router on conditioning statistics; a window in the Swiss Alps and a window on Delhi's edge route to different experts trained on the appropriate morphology. Around the generative core sit the stages a planning office needs to use it: an eleven-metric sustainability scorecard with best-of-N selection, a typed-zoning classifier with per-class abstention, demand forecasting from the same built-up epochs, compliance checking whose thresholds come from jurisdictional rule packs extracted from the planning documents a jurisdiction actually publishes (a zoning ordinance, a comprehensive plan, or a model-code excerpt, read by a deterministic pattern layer and an optional local language model, with every extracted value quoting the sentence it came from), synthesis of public comments into validated plan operations, and deterministic drafting of staff reports in which every sentence traces to a computed quantity. Natural-language control flows through a closed operation schema, and the language model behind it is pluggable, defaulting to a locally served Gemma model so sensitive planning inputs never leave the machine, with a rule parser as the fallback when no model is available. All five experts trained to convergence (final noise-prediction losses 0.024 to 0.042); the zoning classifier reaches leave-one-town-out macro-F1 0.257 across five morphology families; and a controlled reproduction of a previously documented zoning failure on a United States suburb shows the repair came from training data rather than feature calibration, a negative result reported as such. Code, trained weights, a client-side browser deployment, and the full pipeline are public. Duties requiring discretion, negotiation, or legal judgment are identified explicitly and left to people.

1 Introduction

Ask what a machine can contribute to municipal planning and the answer depends on which duty is meant. A planner's week contains comprehensive-plan analysis, development review, zoning administration, demographic forecasting, public engagement, report writing, and a set of political and legal responsibilities that no statistical model should hold. Most generative-urbanism research addresses none of these directly: it asks what a city looks like, produces a plausible layout, and stops. This paper starts from the duties. We take a published inventory of city-planner responsibilities, sort it into what is automatable with current methods, what is assistive at best, and what is human by nature, and then build and evaluate a pipeline that covers the automatable set for one class of problem: where and how a small settlement should grow.

The generative core answers a narrow question: given a particular settlement, on particular terrain, what would its next increment of growth plausibly look like? Three observations motivate the framing. First, most of the world's settlements are small, and their growth decisions are made with limited planning capacity, so tooling helps most exactly where staff is scarcest. Second, terrain binds hardest where capacity is lowest; hill towns in Nepal or the Apennines cannot borrow street patterns from Chicago, yet the most capable published layout generators are trained on large, gridded, mostly American cities [2, 3, 4] and degrade on non-grid morphology [5]. Third, a growth proposal that arrives without evaluation, zoning, compliance findings, and a readable report is not usable by a planning office regardless of its quality; the surrounding stages are not conveniences but the condition of use.

Terrain-aware growth modelling itself is decades old. The SLEUTH family of cellular automata has used slope as a growth driver since the 1990s [1], and learned land-use-change models include elevation among their covariates [6]. Those models output urbanisation probabilities at coarse resolution; they cannot say where the street goes. Diffusion-based layout generators produce street-level designs but condition on land use, text, or nothing, not topography [2, 7, 8], and diffusion work that does engage terrain generates the terrain itself [9, 10]. The combination built here, street-level generative design conditioned on terrain and wrapped in the stages a planning office requires, is to our knowledge new.

The contributions are:

1. A fully automatic data pipeline that assembles terrain-conditioned growth training pairs for arbitrary settlements from open data with no manual annotation, using the 1980–2020 GHS-BUILT-S epochs [21] as real temporal supervision, together with curated training and held-out window lists spanning villages, hilly towns, planned fringes, informal frontiers, and megacity edges across five continents.

2. A five-expert mixture of compact ($\sim$13M parameter) conditional diffusion models sharing one architecture, dispatched by a hard router on interpretable conditioning statistics (slope, built fraction, road fraction, and a roads-per-built ratio that separates unplanned from planned growth), with declared fallback chains for partial deployments and a logged decision for every window.

3. An evaluation and selection layer: an eleven-metric raster scorecard computed against real amenities, greenspace, and terrain hazard proxies, used for best-of-N selection, and a demand-side complement that forecasts built-area growth from the same epochs and scores candidates against the projected budget.

4. A planning-task layer covering duties beyond generation: typed zoning with per-class abstention margins, compliance checking under jurisdictional rule packs extracted from supplied planning documents with quoted evidence and strictest-wins merging, lexicon-based synthesis of public comments into validated plan operations, and deterministic staff-report drafting in which every sentence traces to a computed quantity.

5. A language interface in which planner intent maps onto a closed, validated operation schema, and the model performing that mapping is pluggable behind a one-method interface, default-

ing to a locally served Gemma model with a keyword rule parser as the offline fallback, so control never depends on any single provider and text inputs need not leave the machine.

6. An explicit duty-coverage accounting: which planner responsibilities the pipeline covers, which it assists, and which it does not attempt, published alongside the code so the claim can be audited duty by duty.

2 Related work

2.1 Generative urban layout models

Procedural methods synthesise street networks from grammars and rules [11, 12] at the cost of manual rule-crafting per style. Learning-based successors include autoregressive road-layout generation [13], graph-attention building layout for arbitrary blocks [4], and diffusion over unbounded city layouts [3, 14]. Closest to our setting, He et al. integrate human expertise with multimodal diffusion for urban design, trained on Chicago and New York, and name the restriction to US-grid morphology as a limitation [2]. GAN-based urban form models trained across cities report degraded fidelity on non-grid street patterns [5]. None of these condition on terrain.

2.2 Urban growth and land-use-change models

SLEUTH couples a cellular automaton with slope, land use, exclusion, transport, and hillshade inputs [1] and remains in active application [15]; extensions add support vector machines [16] and deep networks [6], and ConvLSTM models forecast growth from satellite time series [17]. These models simulate whether a cell urbanises, not what is built. Our supervision draws on the same multi-epoch built-up products [21] but uses them as diffusion targets.

2.3 Terrain and geospatial generation

Diffusion models generate terrain from text or sketches [9, 10, 24], and conditioned diffusion synthesises satellite imagery from maps or geospatial primitives [7, 22], with GPS coordinates usable as a control signal [23]. That line treats terrain or imagery as the output. We invert the relationship: terrain is the constraint under which settlement structure must be generated.

3 Growth model

3.1 Problem formulation

Let $E \in \mathbb{R}^{H\times W}$ be an elevation raster over a window of $\sim$1.9 km (H=W=128 at 15 m/px), S its slope, $D_t \in [0,1]^{H\times W}$ the observed built-up density at epoch t, and R_t a binary raster of the road network inside the epoch-t footprint. The model learns $p(R_{\text{new}}, D_{t+1}, A \mid E, S, D_t, R_t, W)$, where R_{new} are roads outside the epoch-t footprint, D_{t+1} is the next-epoch density field, A is a proposed amenity-density field, and W is a filled water mask included as a fifth conditioning channel so that a flat lake does not read as buildable valley floor; training targets are forced to no growth on water. Density is continuous, so infill densification and greenfield sprawl are both supervised and both generable. Elevation is z-scored per window and clipped to $\pm 3\sigma$, slope is normalised by 30°, and targets map affinely to $[-1, 1]$.

3.2 Temporal supervision

GHS-BUILT-S R2023A [21] provides built-up surfaces for the epochs 1975 through 2020. For each window and consecutive epoch pair with sufficient observed growth, one sample conditions on the state at t and targets the state at $t{+}1$; windows without usable change are dropped automatically. One proxy remains and is carried in the limitations: OpenStreetMap has no road history, so epoch-t roads are present-day roads clipped to the epoch-t footprint. A concentric-erosion construction that manufactures pseudo-temporal pairs from a single snapshot, used in the earliest version of this work, is retained as a baseline; its channel counts match, so the same architecture trains on either supervision.

3.3 Amenity channel

The third target channel is amenity density rasterised from OpenStreetMap points of interest, so the model learns where local centres sit relative to fabric and proposes amenity locations for new growth. The channel is calibrated on an absolute scale: blurred, log-scaled counts divided by a fixed reference chosen as the peak a tight cluster of roughly twenty-five points produces through the pipeline's own blur. An earlier per-window maximum normalisation forced some location in every window to read 1.0, with consequences documented in Section 7.

3.4 Five experts and a hard router

A single model trained across growth regimes must average over incompatible morphologies. The pipeline instead trains five experts sharing one architecture: *village* (22 windows around villages with measurable 1980–2020 growth), *town* (hilly small towns of Europe and Asia, the original regime), *urban* (planned flat fringes across Europe, the United States, Australia, New Zealand, and Canada), *informal* (24 unplanned peri-urban frontier windows across Latin America, Africa, and South and Southeast Asia), and *megacity* (24 saturated growth-edge windows, four of them in the Delhi region with a fifth, Narela, held out). Dispatch is a hard, ordered rule on conditioning statistics rather than a learned gate: a built fraction below 0.02 routes to the village expert; above 0.22 to the megacity expert; a built fraction above 0.10 with a roads-per-built ratio below 0.35 to the informal expert, since unplanned frontiers put up buildings faster than mapped roads; a mean slope above threshold to the town expert; and the remaining flat, road-served windows to the urban expert. A hard rule is auditable, cannot silently misroute, and returns the features behind every decision. Deployments rarely carry all five checkpoints, so each expert declares a fallback chain and the dispatcher logs both the requested and the substituted expert. The costs are stated: no blending at threshold boundaries, thresholds that encode our partition of the training data, and a window whose roads exist but are unmapped routing as informal.

3.5 Architecture and training

Each expert is a U-Net with three resolution levels, residual blocks, sinusoidal timestep embeddings, and self-attention at the two coarsest resolutions ($\sim$13M parameters), predicting noise under a cosine schedule with $T{=}1000$ [18, 19]; sampling uses 50 DDIM steps [20]. Training uses AdamW, cosine learning-rate decay, gradient clipping, an EMA of the weights, and dihedral augmentation.

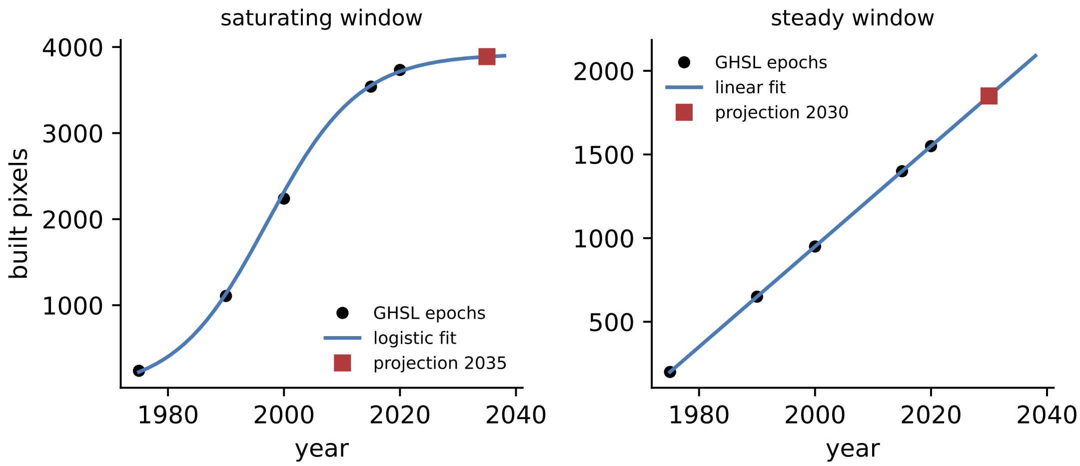

Figure 1: Forecast model selection on the two regimes the stage distinguishes, computed by the released module. Left: a saturating window; the logistic wins the penalised fit and the 2035 projection respects the carrying capacity instead of extrapolating a recent slope. Right: a steadily growing window; the line wins and the 2030 projection continues it.

4 Evaluation, selection, and demand

4.1 Sustainability scorecard

A generative model of plausible growth becomes a planning tool only with a way to compare futures. An eleven-metric raster scorecard grades any (roads, density) plan against real OpenStreetMap amenities in seven categories, greenspace and water masks, and two terrain hazard proxies: 15-minute network amenity coverage by Dijkstra walk times, infill share, land efficiency, circuity, earthwork under new roads, preservation of baseline functional green patches, green access, flood avoidance by a height-above-nearest-water proxy, landslide avoidance above 25°, a congestion proxy from the Gini coefficient of edge betweenness, and access equity as the population-weighted dispersion of coverage. Weights sum to one, hazard terms count new development only, and a growth-delivery factor prevents a do-nothing plan from winning by avoiding every hazard. Best-of-N selection samples N futures, scores each, and keeps the top ranked. The metrics are spatial-form proxies, not certified environmental performance, and socioeconomic factors are named as required human inputs rather than estimated from rasters.

4.2 Demand forecasting and growth budgets

Selection as described rewards plans that deliver growth well, not plans that deliver the right amount of it. A forecasting stage fits the five built-up epochs of each window with a line and a logistic curve, selects by residual error with a complexity penalty on the logistic, refuses horizons beyond twenty years, and converts the projection into a growth budget in pixels. Candidates are then also scored by closeness of delivered growth to the budget, as a rerank feature alongside the scorecard. Figure 1 shows the model selection working as intended on the two regimes it was built for: a saturating series is fitted and capped by the logistic, and a steady series keeps its linear slope. Five points support only rough projections; the output carries its fit residual and a caution that the number is an order of magnitude, and the drafted reports repeat that caution.

5 Annotation stages

Two planning layers stay out of the diffusion targets because their labels are too sparse to supervise generation; both train where labels exist and transfer under leave-one-town-out protocols. A slope-aware logistic classifier over physical edge features marks road segments where protected cycling infrastructure is physically plausible. A gradient-boosted per-pixel classifier types new development as residential, commercial, industrial, institutional, or mixed use, from features the generator already produces; mixed use is labelled by proximity co-occurrence of residential and commercial pixels, since OpenStreetMap carries no first-class mixed-use tag at this resolution. The classifier abstains rather than guesses: a pixel whose classification margin or local density context falls below threshold stays untyped, and per-class margins let a deployment demand more evidence for a designated class without touching the others. An advisory pass types the undeveloped ring around the existing footprint before any sample commits growth there, drawn faint in the deployment so recommendation and proposal cannot be confused.

6 The planning-task layer

This section covers the duties beyond generation. The design rule throughout is that language routes and validates but never draws: geometry comes only from a fixed region grammar, classes and metrics are checked against closed vocabularies, and everything still flows through the same scorecard.

6.1 Language operations and pluggable models

A planner's sentence such as "more commercial near the centre and protect the floodplain" maps onto a short list of verified operations: Regenerate a region, SetZoneBias, AddAmenityTarget, Protect, or Rerank the scorecard weights. Regions resolve through a fixed mask grammar; unknown clauses become warnings, never guessed operations; Protect and Regenerate execute through RePaint-style region locking, in which locked pixels are re-injected at the current noise level at every reverse step and composited exactly at the end.

The model that performs the sentence-to-operations mapping is pluggable behind a one-method interface. Three backends ship: a local Ollama server whose default model is Gemma, chosen because it runs on commodity hardware under a permissive license and keeps resident comments and unpublished drafts on the machine; any OpenAI-compatible endpoint, which covers hosted providers and local servers exposing that interface; and a deterministic offline backend used by tests and as the automatic fallback, alongside the keyword rule parser, when no model is reachable. Every backend's output passes the same strict schema validation, so swapping models cannot widen what the system is permitted to do.

6.2 Public comment synthesis

Planning departments receive written comments and someone must count who supports what and which requests recur. A lexicon-based stage tags each comment with a stance (support, oppose, concern, neutral) and topics, aggregates the batch, and converts requests raised by at least a configurable number of distinct comments into validated operations from the same closed schema, so public input steers the same tools a planner steers. Lexicon methods are inspectable and cheap, and their limits are stated in the output itself: clauses containing negation markers are dropped from operation extraction with a warning, copy-pasted duplicates count once, and the packet section says plainly that automated tagging misses sarcasm and non-English comments and that a staff read of the full record remains required.

6.3 Jurisdictional rule packs

No single set of compliance thresholds is the law anywhere, so the applicable rules are an input rather than a constant, and the pipeline obtains them by reading the documents a jurisdiction actually publishes: a zoning ordinance, a comprehensive plan, a hillside or floodplain chapter, or an excerpt of a model code such as the International Building Code. No code text is bundled with the pipeline; model codes are copyrighted and local amendments control in any case, so the caller supplies the text and the extractor works from exactly what it was given.

Extraction runs in two layers over the same text. A deterministic sentence-level extractor recognises the recurring numeric patterns of planning English: slopes exceeding a stated percentage or degree figure, required separations between industrial and residential uses, contiguity requirements that cap the distance of new development from existing services, and floodplain permit language. It works offline and its patterns are inspectable. A language model, reached through the same pluggable backend as the operation parser, reads the document chunk by chunk against a fixed JSON schema over the same closed set of rule keys and proposes further candidates. Both layers emit candidates, never conclusions. Every candidate must carry the verbatim sentence it came from; candidates merge under a strictest-wins rule (the lowest slope limit, the widest separation, the tightest contiguity distance); merged values pass the same range validation as hand-written packs, and a value outside its plausible range is dropped with a warning rather than clamped, since a clamped number would misquote the document. The result is a rule pack: a small data file naming its source document, quoting the provisions it read, marked at the "extracted" level, and stating in its own verification text that every value must be confirmed against the document before reliance. An extracted number is a reading aid; the document remains the authority.

Every findings list produced through a pack ends with a provenance note naming the pack, its level, its quoted evidence, and its verification statement, so a report cannot silently convert a machine reading into a legal determination. When a pack marks flood areas as regulated, growth on mapped water is escalated from a hazard note to a permit matter, and a missing water mask is itself reported as an unchecked regulated layer. Hand-written packs remain supported through the same schema at a "manual" level, which requires citations and a verification statement just as extracted packs do; the schema rejects uncited jurisdictional claims and out-of-range thresholds regardless of how the pack was produced.

6.4 Compliance checking

At 15 m per pixel there are no setbacks or parking counts, so the checks cover what raster geometry supports: growth on water or protected land and on slopes past the applicable limit (violations, since the pipeline's own locks should make them impossible), residential pixels within the separation buffer of industrial typing and leapfrog development beyond the service ring (cautions), and intensity or abstention patterns worth a look (notes). Thresholds are visible constants or pack values, never learned quantities. Non-finite inputs are rejected rather than silently passed, after review found that a no-data window would otherwise read as compliant.

6.5 Staff report drafting

The staff report is the document a hearing body reads. A deterministic drafting stage assembles one from the pipeline's artifacts: site header, the routed expert, scored alternatives with their strongest and weakest metrics, land-use composition, the growth outlook with its caution, the public-comment section, compliance findings, and a recommendation produced by a disclosed mechanical rule (any violation forces denial pending correction; cautions become conditions; a clean report yields approval). Free-form generation was rejected because a reviewable draft must trace every sentence to a quantity that appears elsewhere in the packet. The draft states on its face that it is machine-produced and requires planner review and signature, and the emitted text is checked against a banned-construction list before it is returned. The trade-off is stated: deterministic templates read plainly and cannot adapt tone; adaptation is exactly what reviewability forbids here.

6.6 Duty coverage

The repository publishes a duty-by-duty accounting derived from published planner job descriptions, sorted into three classes. Covered, by the stages above: growth alternative generation, plan evaluation and selection, zoning annotation, demand forecasting, raster-scale compliance review, public comment synthesis, and staff report drafting. Assisted, where the pipeline contributes but a person does most of the work: comprehensive plan updates, site suitability studies, and grant or capital documentation that can quote pipeline outputs. Not attempted, by design: discretionary approvals, variance and appeals judgment, inter-agency and political negotiation, public meeting facilitation, legal interpretation, and enforcement. The accounting is a claim about task types, not headcount, and the assisted class in particular reflects our judgment rather than a measurement.

7 Results

7.1 Training

All five experts trained to 300 epochs on a single T4 GPU with no divergence. Figure 2 shows the five per-epoch loss histories; the final noise-prediction losses order by morphological complexity (village 0.024, town 0.028, urban 0.032, informal 0.038, megacity 0.042), and the ordering holds across most of training, not only at the end. An earlier single-model run on the temporal pairs converged from 0.744 to $\sim$0.040 over 120 epochs with the same profile; training loss is evidence of learnability, not a fidelity claim. On the five held-out windows (Wengen, Gubbio, Des Moines, Antananarivo, Delhi-Narela) the router selected the intended expert in every case. Not every channel learned: the road channel, supervised as a 1-pixel line, does not reach the spatial coherence of the density and amenity channels (neighbour correlation $\sim$0.26 against $\sim$0.98), the deployment hides it, and road-target dilation is the identified but untested fix.

7.2 Selection behaviour

Figure 3 shows best-of-N selection end to end on a held-out town: the top three futures score within one point of each other (51, 50, 50 of 100) while proposing visibly different arrangements, with 15-minute amenity coverage between 82% and 87%. This is the intended behaviour: several near-equivalent options for a person to choose among rather than a single answer.

7.3 Zoning classifier and a controlled negative result

The typed-zoning classifier reaches leave-one-town-out macro-F1 0.257 on the five-family training distribution, below the $\sim$0.30 of its predecessor trained on hilly European towns alone; transfer

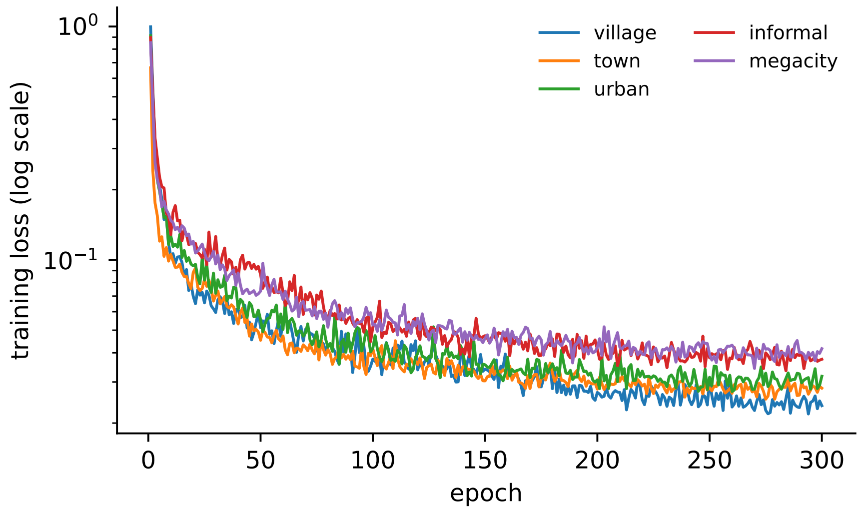

Figure 2: Per-epoch noise-prediction loss (log scale) for the five experts over 300 epochs, from the recorded training histories. The curves separate by morphological complexity early and keep that order: the village expert converges lowest and the megacity expert highest.

across five morphology families is the harder problem, and abstention exists because a low-F1 classifier should indicate rather than assert.

A deployment observation motivated a controlled experiment. On a window over Middleburg Heights, Ohio, an earlier classifier labelled 697 of 1,099 generated-growth cells and 4,869 of 11,345 advisory cells institutional, traceable to per-window maximum normalisation of the amenity feature: a single school on a parking lot became the window's maximum, and the classifier had never seen an isolated, unclustered amenity peak. The reproduction protocol retrained two classifiers on the expanded training distribution, one under the old normalisation as a control and one under the absolute calibration. The failure did not reproduce under either: institutional counts were 203 and 221 of roughly 4,300 typed advisory cells respectively, both near 5% against the original 43% (Figure 4, left), and macro-F1 was identical to three decimal places. The repair therefore came from the training data, not the feature calibration, which is retained on measurement-consistency grounds. The per-class margin behaves monotonically, with institutional assignments falling 221, 104, 32, 1 as the margin rises through 0, 0.5, 1.0, 2.0 and every other class unchanged (Figure 4, right); 0.5 is the deployed default. The dense in-distribution check did not regress: Gubbio's advisory typing is identical under every variant. What the protocol cannot isolate is the originally deployed classifier itself, whose cache predates the expanded lists; the 43% figure rests on the deployment observation.

7.4 Deployment

The full stack runs client-side in a browser: each expert exports to a 23 MB ONNX file fetched only when the router selects it, the zoning classifier evaluates as JSON decision trees in JavaScript (reproducing the Python classifier's predictions exactly where checked), and a real-town mode fetches live elevation and OpenStreetMap data for any clicked location. Figure 5 shows two Rhine-valley windows outside every training and evaluation list. The inference-time locks act exactly at their boundaries, with no growth or typing on the river channel or the scarp; the bike classifier marks much of the flat grid, its expected behaviour given training on flat cycling towns; most colour in the built-out Vaduz window is the faint advisory layer over expansion land rather than committed growth; and the advisory typing between the town core and the river reads industrial, which coincides with Vaduz's actual industrial estates. Two caveats accompany the images: each is a single deterministic sample, and the display thresholds separating interior from expansion land were calibrated against the Vaduz window, so the modest committed-growth footprint there is partly a property of the calibration.

7.5 Verification of the planning-task layer

The planning-task modules were subjected to an adversarial review before release. The review produced twenty defects across the four task modules, four of them of the highest severity, including an import-time failure on Python versions below 3.12, a crash on operations containing dictionary-valued fields, inversion of negated public comments into their opposite operations, and report generation crashing on place names containing banned stems. All twenty were corrected, each correction is covered by a regression test in the module's self-test suite, and all suites pass offline. The defect list and fixes are in the repository history.

8 Limitations

Temporal road proxy. Epoch-t roads are present-day roads clipped to the epoch-t footprint, so recent roads leak into early conditioning; the built-up channels are genuinely temporal. **Density granularity.** GHS-BUILT-S is native $\sim$100 m resampled to 15 m/px; the density target cannot carry parcel-level structure. **Road channel.** Unlearnable at this data scale as supervised; hidden in deployment; dilation untested. **Label sparsity.** OpenStreetMap coverage thins outside Europe; road mapping is thin in several informal windows and building mapping in the Chinese megacity windows, so the growth filter drops some of exactly the windows those experts most need. **No mixture-versus-single comparison.** All five experts are trained, but no controlled comparison against a single model on pooled data has been run; the case for the mixture rests on the training-distribution argument. **Router thresholds.** Chosen from training-window distributions, not validated against an external definition of settlement type. **Zoning accuracy.** Macro-F1 0.257; abstention reduces the cost of errors, not their rate. The farmland class exists in the pipeline but received no surviving training labels and never fires. **Forecasts.** Five points per window; the stage refuses long horizons and labels its outputs as order-of-magnitude, and both properties are load-bearing. **Comment synthesis.** A lexicon; sarcasm, cross-clause negation, and non-English comments defeat it, and the output says so. **Rule extraction.** The deterministic extractor reads only the sentence patterns it was written for, and its recall and precision on a corpus of real ordinances have not been measured; the language-model layer inherits whatever reading errors its backend makes, which is why both layers must quote their evidence and why extracted packs demand confirmation against the document. A quoted sentence is not an interpretation: definitions, exceptions, overlays, and amendments elsewhere in a code can change what a number means, a raster mask is not a flood map, and a distance ring is not a zone plan. No extracted pack has been reviewed by an authority of the jurisdiction whose document it read. **Report drafting.** Deterministic templates trade fluency for traceability; the recommendation rule is mechanical and binds no one. **Scorecard scope.** Spatial-form proxies; no zoning law, land ownership, budget, or hydraulic model enters the pipeline, and generated layouts are proposals. **Held-out evaluation.** The slope-occupancy, connectivity, contiguity, and volume protocol is implemented and shipped, but the systematic held-out run across all windows has not been completed.

9 Conclusion

Starting from what city planners actually do, this paper assembled and evaluated a pipeline that covers the automatable subset of those duties for settlement growth: five terrain-conditioned diffusion experts routed by an auditable rule, scorecard evaluation with demand-aware selection, abstaining zoning annotation, compliance under rule packs extracted with quoted evidence from the planning documents jurisdictions publish, public comment synthesis into a closed operation schema, deterministic staff-report drafting, and a pluggable language interface defaulting to a locally served model. The measured results include converged training across five growth regimes, correct routing on all held-out windows, a zoning classifier whose accuracy is reported at 0.257 rather than dressed up, and a controlled negative result attributing a documented failure's repair to training data rather than feature engineering. The duties left out are left out on purpose. Discretion, negotiation, and legal judgment belong to people; the pipeline's job is to make the paperwork around those judgments faster and the analysis under them reproducible.

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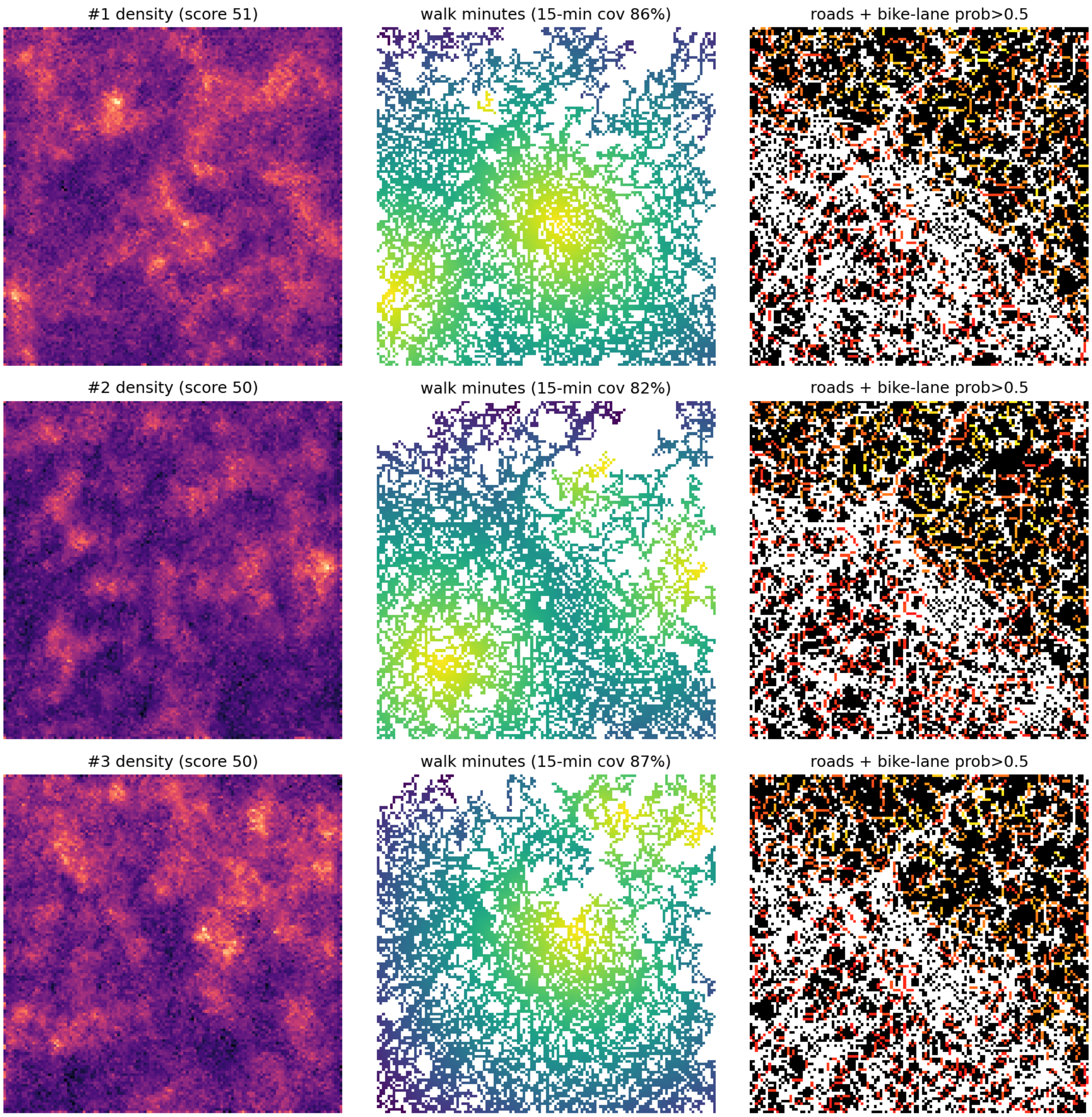

Figure 3: Top-3 of N sampled futures for a held-out town under the scorecard (totals 51, 50, 50 of 100). Left: generated continuous density. Centre: per-pixel walk time along the generated network to the nearest real amenity, with 15-minute coverage per sample (86%, 82%, 87%). Right: road pixels with edges whose predicted bike-infrastructure probability exceeds 0.5 overlaid. Raw model rasters, shown without the rendering stage.

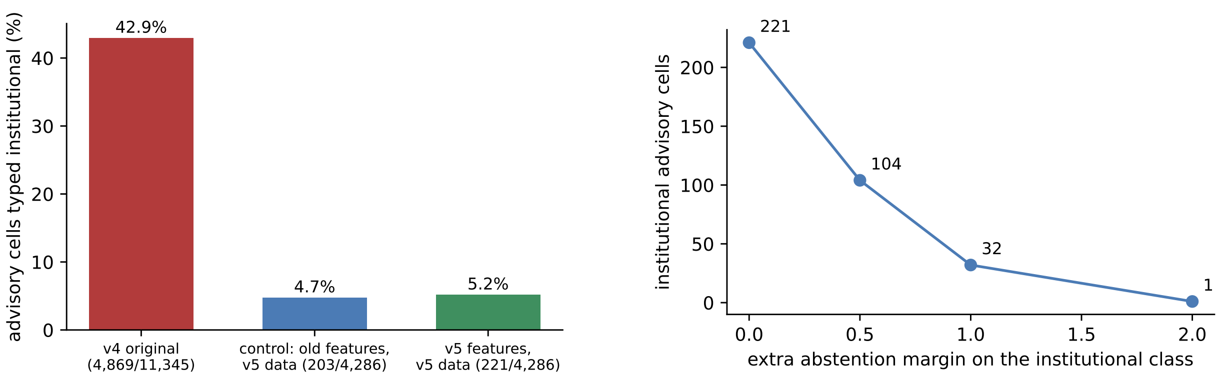

Figure 4: The Middleburg Heights experiment. Left: share of advisory cells typed institutional in the original deployment observation and in the two retrained classifiers (old features as control, absolute amenity calibration); the control repairs the failure on its own, which is what makes the result negative for the calibration hypothesis. Right: institutional advisory count under the per-class abstention margin sweep; every other class is unchanged at every setting.

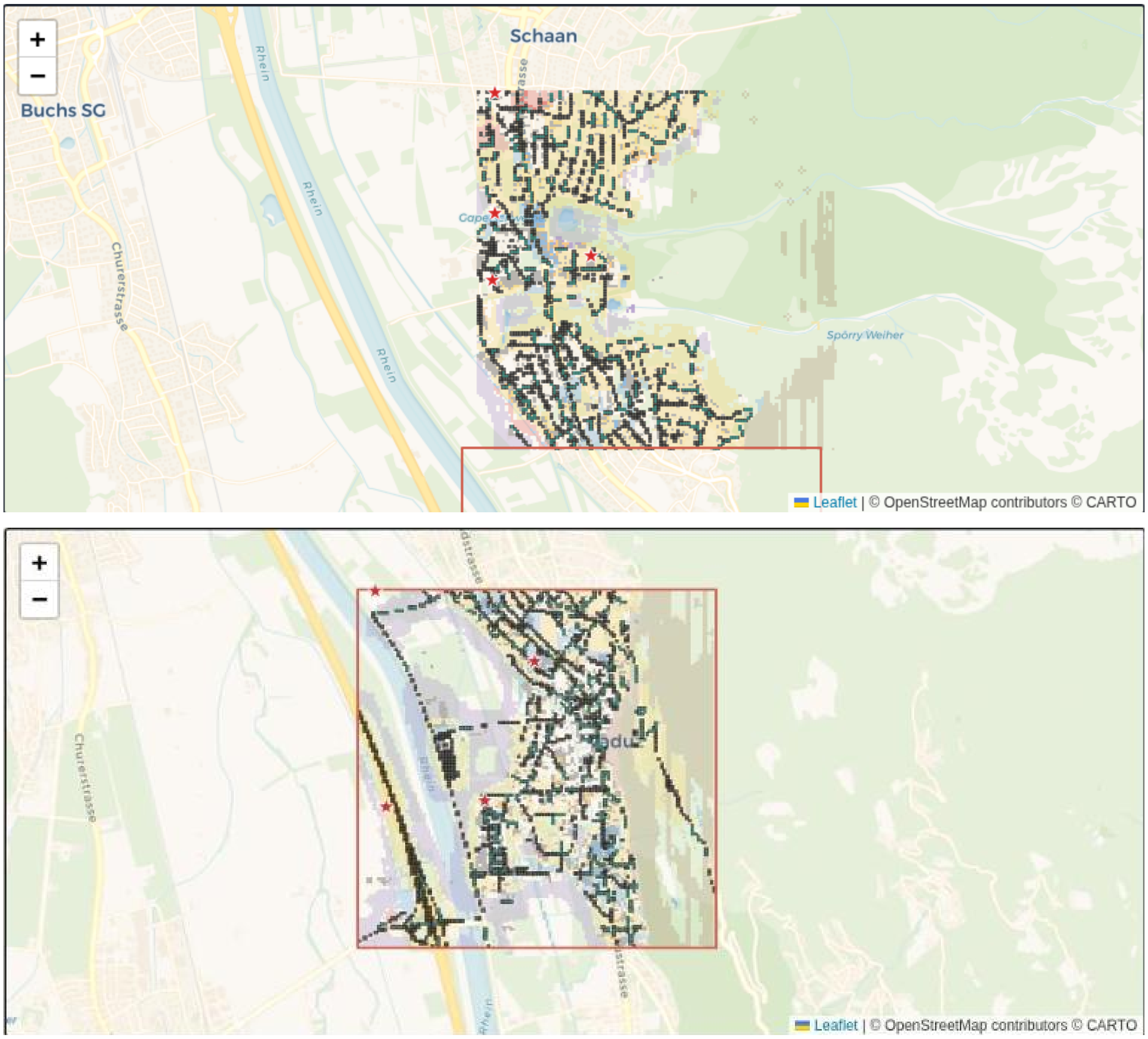

Figure 5: The deployed browser tool on two windows outside all training and evaluation lists: Schaan (top) and Vaduz (bottom), Liechtenstein. Dark pixels are the existing road network; teal marks segments with predicted bike-infrastructure probability above 0.5; solid colours are sampled growth typed by the zoning classifier; faint colours are the advisory layer; stars are proposed local centres. Growth and typing stop at the water lock on the Rhine and the steep-slope lock on the scarp. Single deterministic samples; see the text for caveats.